

Q2. Add-1 Smoothing (based on slide: Worked Sentiment Example)

In the worked example, priors are: $P(-)=3/5$, $P(+)=2/5$. Vocabulary size = 20.

Tasks:

$$p(w_i|c) \models \frac{\text{count}(w_i, c) + 1}{(\sum_{w \in V} \text{count}(w, c)) + |V|}$$

1. For the negative class, the total token count is 14. Compute the denominator for likelihood estimation using add-1 smoothing.

We are told:

total tokens in negative docs = 14

vocabulary size = 20

Denominator 14+20 = 34

2. Compute $P(\text{predictable}|-)$ if the word “predictable” occurs 2 times in the negative documents.

Count of “predictable” in negative docs = 2

$P(\text{predictable}|-) = 2+1/34 \approx 0.088$

3. Compute $P(\text{fun}|-)$ if “fun” never appeared in any negative documents.

Count of “fun” in negative docs = 0

$P(\text{fun}|-) = 0+1/34 \approx 0.029$